Design of a D-PHY Chip for Mobile Display Interface Supporting MIPI Standard

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Abstract-- **This paper presents a D-PHY chip design for MIPI (Mobile Industry Processor Interface) standard. The MIPI is a flexible, source-synchronous serial interface standard connecting a host processor to a display and camera modules as used in mobile devices. The D-PHY consists of LP (low-power) mode block, HS (high-speed) mode block and control blocks. We implemented D-PHY chip using 0.13-um CMOS process under 1.2V supply. As a result, HS mode shows 1Gbps with jitter 5% and 0.74mW power consumption.**

I. INTRODUCTION

MIPI is a synchronous serial interface standard between the host processor and peripherals such as display module, camera module typically found in mobile display system. The D-PHY link supports a high-speed (HS) mode for fast data transfer and also a low-power (LP) mode for control transactions. Fig.1 shows the MIPI D-PHY lane module that consists of a digital block for controls and an analog block for data transmission. The lane module has a transceiver portion handles differential HS functions (HS-TX, HS-RX) utilizing both interconnect wires simultaneously and single-ended LP functions (LP-TX, LP-RXs) operating on each of the interconnect wires individually. This paper presents design and simulation of low power D-PHY lane module [1]-[3].

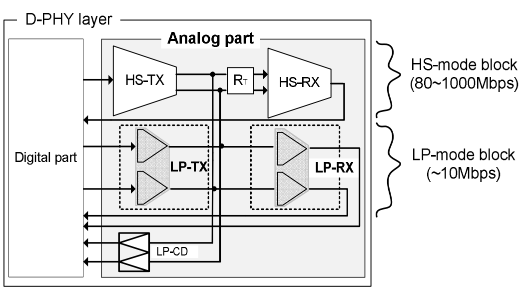


Fig. 1. Diagram of MIPI D-PHY lane modules.

II. DESIGN AND SIMULATION OF MIPI D-PHY

The D-PHY digital block controls transferring and receiving of image data and commands for control. The transmitter sends out the serialized packet data is organized by the upper protocol layer. The receiver changes format of the received serial packets to parallel. Fig. 2 shows the signalings of each mode in MIPI D-PHY. The HS mode supports synchronous NRZ HS data transmission using SLVS technique. The HS-TX transmits data at a small swing, $0.2V_{pp}$, differential signal with common-mode voltage, 0.2V. The signaling pair has a bit rate 80Mbps up to 1000Mbps. In HS mode, each transmission lane is terminated on both sides. The LP functions are mainly used for the controls and asynchronous LPDT (low power data transmission). The single-ended LP functions are un-terminated and always presented in pairs. LP mode has to support full swing of data due to LVCMOS signaling, e.g. 1.2V within 10Mbps. When no data is transmitted, both TX and RX may shut off to conserve power consumption.

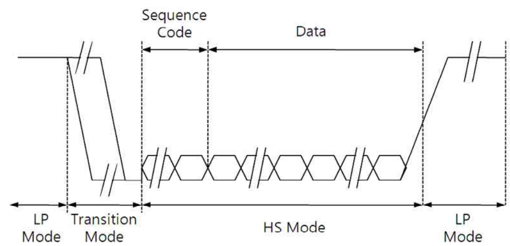
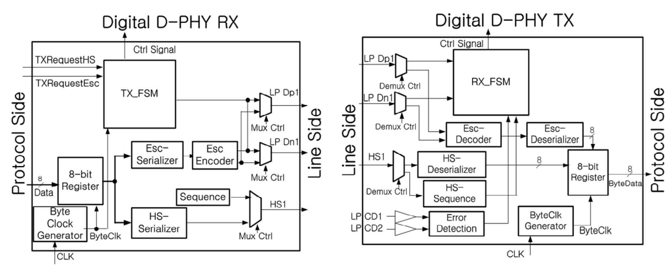


Fig. 2. Signal pattern of HS mode and LP mode.

Fig. 3 shows the digital block of the transmitter and receiver. In left side of transmitter is the input signal from the upper protocol layer. The output of TX and input of RX are analog interface blocks.



(a) (b)

Fig. 3. Block diagram of Digital D-PHY. (a) Transmitter, (b) Receiver

Fig. 4 shows LP mode circuit diagram. The CMOS drivers were divided into three pieces as simultaneous switching output, noise reduction and slew-rate control. Fig. 4(a) controls slew-rate and limits the output current choosing the output buffers with a push-pull driver [4] to keep EMI low. The circuit has 3 inverter type drivers, MND1+MPD1, MND2+MPD2, MND3+MPD3, with different size for different driving capacity. As shown in Fig. 4(b), the proposed accurate LP-RX consists of two hysteresis comparators, inverters, and latche. The input for high and low level voltages are v_{ref1} and v_{ref2}.

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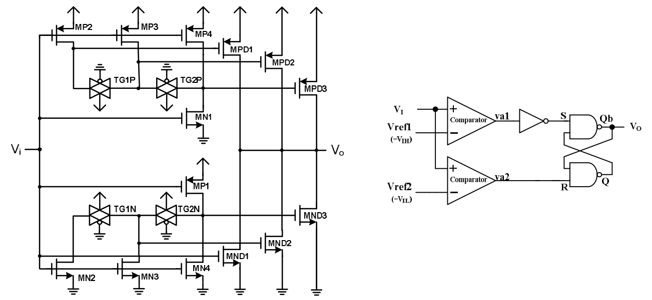


(a) (b)

Fig. 4. Schematics of the LP mode circuits. (a) LP-TX, (b) LP-RX

In Fig. 5, HS-TX supports synchronous differential high-speed data transmission based on SLVS [5]. The SLVS is a chip-to-chip signaling protocol that uses differential voltage-mode signaling. The HS-TX transmits $0.4V_{pp}$ differential voltage with nominal $0.2V_{pp}$ single-line swing and 0.2V common mode voltage.

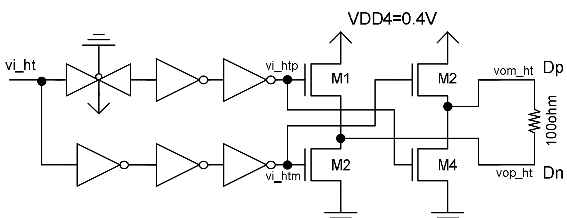


Fig. 5. Schematics of the HS-TX circuit.

The proposed HS-RX in Fig. 6 consists of a termination resistor (ZID), CMOS differential amplifiers VCDA (very-wide-common mode-range differential amplifier), OTA and buffer stage. First stage is VCDA suppressing the fluctuations of ΔV_{CMRX} and ΔV_{OD}. Second stage is single-ended OTA amplifies the outputs of VCDA stable. The last stage is a buffer using multi-stage inverters that controls power consumption for load capacitances.

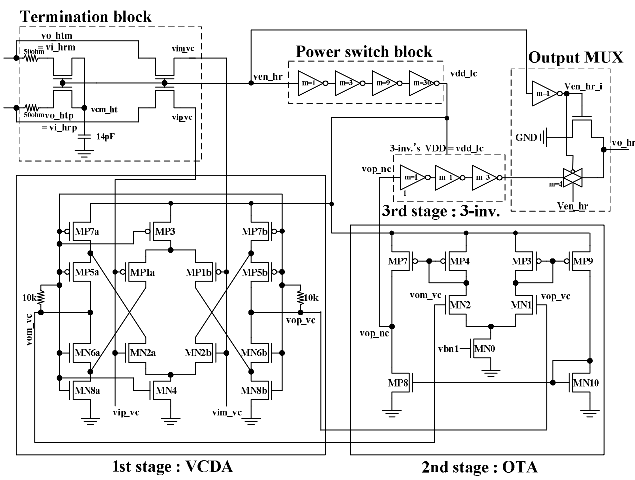


Fig. 6. Schematics of the HS-RX circuit.

The simulation results of LP-TX with various load capacitances are shown in Fig. 7. The maximum current of LP_TX is limited by slew-rate control that the transition times of LP-TX outputs meet with the specification.

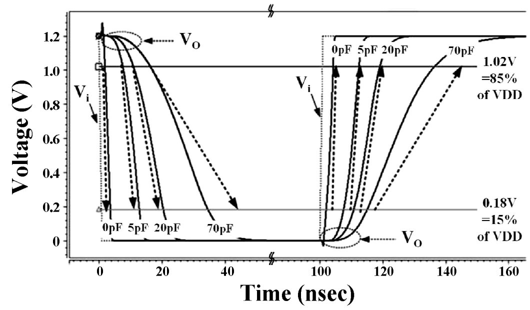
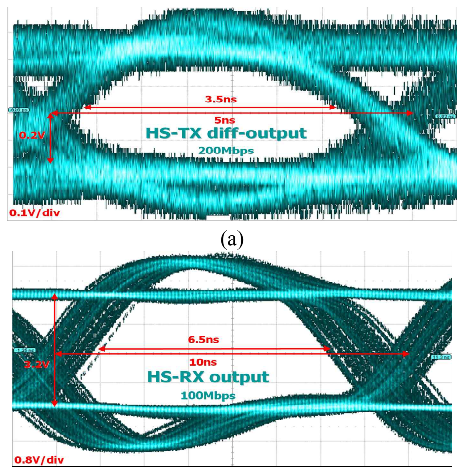


Fig. 7. Simulation results of LP-TX with load capacitance variation.

Fig. 8 shows the measured eye patterns of MIPI D-PHY chip with a 2^{10}-1 PRBS input data. Fig. 8(a) shows an eye pattern of HS-TX differential output (V_{Dp} - V_{Dn}) at 200Mbps. The eye opening and width of HS-TX are $0.3V_{pp}$ and 3.5ns, respectively. Fig. 8(b) is an eye pattern of HS-RX output at 100Mbps and the jitter of 3.5ns.



(a)

(b)

Fig. 8. Measured eye pattern of MIPI D-PHY chip (a) HS-TX, (b) HS-RX

III. Conclusions

We designed the MIPI D-PHY chip supporting both HS and LP mode content with MIPI standard. The D-PHY analog part consists of HS mode blocks, LP mode blocks and control block. We implemented MIPI D-PHY chip using 0.13-um CMOS process under 1.2V supply. As the results, TX and RX at HS mode shows data transmission speed up to 400Mbps and 500Mbps consuming power with 0.52mW and 0.74mW with at jitter 5%, respectively.

References

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